CLASS 8 — MATHS · FORMULA SHEET

Chapter 5: **Number Play** | Quick revision, one page

Divisibility tests

- By 2: units digit 0, 2, 4, 6, 8. By 5: units digit 0 or 5. By 10: units digit 0.
- By 3: digit sum divisible by 3. By 9: digit sum divisible by 9.
- By 4: last two digits divisible by 4. By 8: last three digits divisible by 8.
- By 6: divisible by both 2 and 3.
- By 11: (sum of digits at odd places) $-$ (sum of digits at even places) is 0 or a multiple of 11.

Place-value expansions

- $\overline{ab} = 10a + b$ and its reverse is $10b + a$; $\overline{ab} + \overline{ba} = 11(a + b)$.
- $\overline{abc} = 100a + 10b + c$; $\overline{abba} = 11(91a + 10b)$.

Number patterns

- Palindrome: a number that reads the same forwards and backwards (e.g. 1331).
- Kaprekar's routine (largest arrangement $-$ smallest arrangement): 3-digit numbers (not all digits equal) reach 495; 4-digit numbers reach 6174.
- A number and the sum of its digits leave the same remainder on division by 9.

Rules to use

- If a number is divisible by a and b (co-prime), it is divisible by ab.
- The sum of any 3 consecutive numbers is divisible by 3; the sum of any n consecutive numbers starting at m is $nm + \dfrac{n(n-1)}{2}$.

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